

The term **migration** can mean different things depending on the context. In information technology (especially cloud computing, virtualization, and data centers), migration generally means **moving data, applications, services, or entire systems from one environment to another**.

What is Migration?

Migration is the process of transferring something from a source system to a target system.

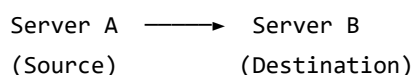
Examples:

- **Data migration:** Moving data from one database to another.
- **Application migration:** Moving an application from one server or cloud provider to another.
- **Virtual Machine (VM) migration:** Moving a running virtual machine from one physical host to another.
- **Cloud migration:** Moving workloads from on-premises infrastructure to cloud platforms.

Basic Migration Process

1. Prepare the destination system
2. Copy data or system state
3. Synchronize any changes
4. Switch users/services to the new location
5. Verify everything works correctly

Example:



What is Live Migration?

Live migration is a technique that moves a **running virtual machine (VM)** or workload from one physical host to another **without shutting it down and with minimal or no service interruption**.

It is commonly used in virtualization platforms such as:

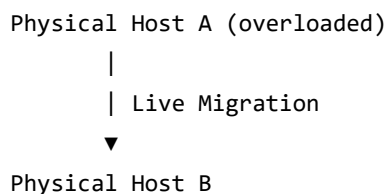
- VMware vSphere
- Microsoft Hyper-V
- KVM
- Xen

Why Use Live Migration?

Administrators perform live migration to:

- Maintain hardware without downtime.
- Balance workloads across servers.
- Avoid overloaded hosts.
- Save power by consolidating workloads.
- Recover from potential hardware failures.

Example:



Users typically do not notice the move.

How Live Migration Works

A VM consists of:

- CPU state
- Memory (RAM)
- Disk storage
- Network connections
- Device state

The challenge is moving all of this while the VM continues running.

The most common method is called **Pre-Copy Live Migration**.

Step 1: Start VM on Source Host

Host A
└─ VM Running

The VM continues serving users.

Step 2: Copy Memory Pages

The hypervisor begins copying memory pages from Host A to Host B.

Host A RAM → Host B RAM

While copying, the VM is still running.

Step 3: Track Changed Pages

Some memory pages change while they are being copied.

These are called **dirty pages**.

Example:

Page 1 copied
Page 2 copied
Page 3 modified again ← dirty page

The hypervisor tracks these modifications.

Step 4: Re-copy Dirty Pages

The system repeatedly copies the pages that changed.

Round 1: Copy all pages
Round 2: Copy changed pages
Round 3: Copy newly changed pages

Eventually only a small number of pages remain dirty.

Step 5: Brief Pause

The VM is paused for a very short time (often milliseconds).

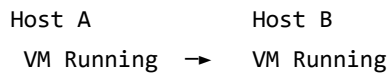
Pause VM

During this pause:

- Remaining dirty pages are copied.
- CPU registers are transferred.
- Device state is transferred.

Step 6: Resume on Destination

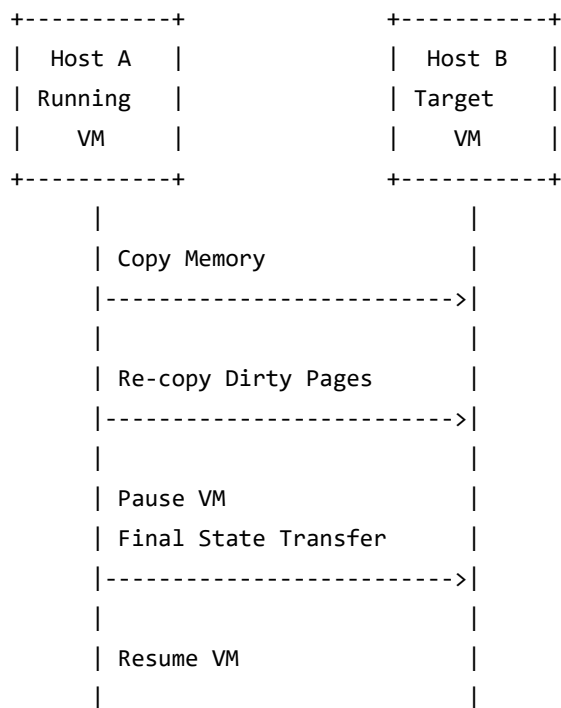
The VM starts running on Host B.



Network traffic is redirected to the new host.

Users typically experience little or no interruption.

Pre-Copy Live Migration Diagram



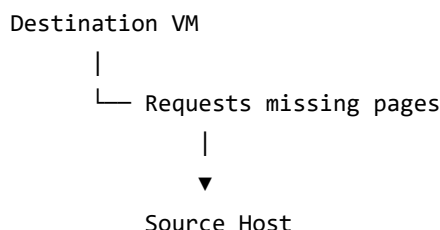
Post-Copy Live Migration

Another approach is **Post-Copy Migration**.

Instead of copying all memory first:

1. Pause VM.
2. Transfer CPU state.

3. Start VM on destination.
4. Fetch memory pages from source as needed.



Advantages

- Faster migration.
- Less duplicate copying.

Disadvantages

- More sensitive to network failures.
- Higher complexity.

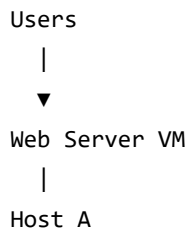
Live Migration Requirements

Typically, the source and destination hosts need:

- Compatible CPUs.
- Shared or accessible storage.
- Reliable high-speed network.
- Hypervisor support.
- Sufficient RAM and resources on the destination host.

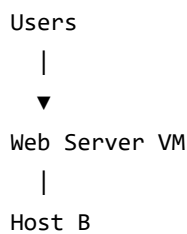
Example

Suppose a web server VM is running on Host A:



The administrator wants to update Host A.

Using live migration:



The VM moves to Host B while users remain connected and the website stays available.

Summary

- **Migration** = moving data, applications, or systems from one environment to another.
- **Live migration** = moving a running VM or workload to another host with little or no downtime.
- The most common technique is **pre-copy migration**, where memory is copied while the VM continues running, followed by a brief pause to transfer the final state.
- Live migration is widely used in virtualized and cloud environments to perform maintenance, load balancing, and fault avoidance without disrupting users.